

FINITE-HORIZON BLOCK AND SLIDING-WINDOW RUIN FOR BROWNIAN RISK MODELS

KRZYSZTOF DĘBICKI, ENKELEJD HASHORVA, AND ZBIGNIEW MICHNA

ABSTRACT. We obtain exact high-level asymptotics for block and sliding-window ruin over a fixed horizon in the Brownian risk model. Block ruin requires an exceedance in each reporting block; sliding-window ruin requires one in every window of the same length. A first-passage argument separates the first-block Gaussian cost from a finite-level crossing or Parisian-type survival probability. We also establish simultaneous block-ruin asymptotics for correlated bivariate Brownian risks in the binding regime. For a compound-Poisson risk model perturbed by Brownian motion, non-negative subexponential claims give the same block and sliding asymptotic, determined by the first reporting period.

1. INTRODUCTION

Let $B = \{B(t), t \geq 0\}$ be standard Brownian motion. The Brownian risk model has reserve $R_u(t) = u + ct - B(t)$, where $u > 0$ is the initial reserve and $c > 0$ the premium rate. We work with the corresponding loss process

$$X(t) = B(t) - ct, \quad t \geq 0,$$

so that $X(t) > u$ means ruin at time t .

The classical finite-time ruin probability

$$\psi(u, T) := \mathbb{P} \left\{ \sup_{0 \leq t \leq T} X(t) > u \right\}$$

records whether the reserve becomes negative at least once before T . To distinguish an isolated crossing from distress recurring across monitoring periods, we require exceedances over prescribed reporting blocks or sliding windows.

Throughout, fix $h > 0$ and an integer $m \geq 2$, and put $T = mh$. All model parameters are fixed as $u \rightarrow \infty$. The *block-window ruin probability* is

$$(1) \quad p_h^{\text{bl}}(u, T) := \mathbb{P} \left\{ \bigcap_{k=1}^m \left\{ \sup_{t \in [(k-1)h, kh]} X(t) > u \right\} \right\}.$$

Thus ruin is required to occur at least once in each of the m reporting blocks. The related *sliding-window ruin probability* is

$$(2) \quad p_h^{\text{sl}}(u, T) := \mathbb{P} \left\{ \inf_{t \in [0, T-h]} \sup_{s \in [t, t+h]} X(s) > u \right\}.$$

For continuous paths, the moving-window maximum is continuous in the window start. Thus (2) means that every length- h window contains a ruin epoch, or equivalently that there is no recovery interval of length h on which $X \leq u$.

Date: September 19, 2026.

Key words and phrases. Brownian risk model; perturbed Brownian risk model; compound Poisson claims; subexponential distribution; block ruin; simultaneous block ruin; bivariate Brownian risk model; sliding-window ruin; finite-time ruin; exact asymptotics.

The sliding requirement is related to, but distinct from, classical Parisian ruin, which requires the reserve to remain below zero throughout an interval of prescribed length. Parisian ruin for spectrally negative Lévy risk processes is studied in [2, 13]; Gaussian counterparts for self-similar models and over finite horizons are treated in [3, 4]. Discrete monitoring of Brownian ruin, including Parisian variants, is studied in [12], while [14] treats Parisian ruin for locally self-similar Gaussian processes. Our sliding event instead excludes sustained recovery; its post-hit constant has the Parisian-type survival representation (11).

Writing Ψ for the standard normal upper tail, our main results give

$$p_h^{\text{bl}}(u, mh) \sim 2G_m(h, c)\Psi\left(\frac{u + ch}{\sqrt{h}}\right), \quad p_h^{\text{sl}}(u, mh) \sim 2G_m^{\text{sl}}(h, c)\Psi\left(\frac{u + ch}{\sqrt{h}}\right).$$

Conditional on a first-block exceedance, the first hit of u approaches h . The strong Markov property at this hit reduces the remaining requirements to finite-level Brownian events, giving the constants G_m and G_m^{sl} . The sliding argument also yields an exact one-dimensional integral representation.

Related simultaneous-crossing problems belong to the theory of Gaussian conjunctions. Exact asymptotics for finite-horizon conjunctions of independent Gaussian coordinates are obtained in [6]. For correlated Brownian risks, infinite-horizon orthant hitting is studied in [5], and finite-horizon bivariate simultaneous ruin in [7]. Finite-horizon simultaneous ruin for classes of multivariate Gaussian risk models with stationary increments is treated in [1]. Related simultaneous-ruin asymptotics for independent gamma-reflected Brownian motions are obtained in [11]. These events require the coordinates to cross their respective levels at a common time; our block event requires such a simultaneous crossing in every reporting block.

We then consider two extensions. For bivariate Brownian risks with correlation ρ and thresholds u, au , we treat simultaneous block ruin in the binding regime $\rho < a \leq 1$. The first-block contribution is the bivariate finite-time ruin asymptotic, while later blocks give a finite-level simultaneous-crossing factor. For a compound-Poisson risk model perturbed by Brownian motion, with non-negative subexponential claims, both repeated-ruin probabilities are asymptotic to $\lambda h \bar{F}(u)$, because one sufficiently large first-block claim dominates. Proofs are collected in Section 4.

2. MAIN RESULTS

In this section, $X(t) = B(t) - ct$, $c > 0$, with the fixed h, m, T specified above. Let

$$\Psi(x) := \mathbb{P}\{Z > x\}, \quad \varphi(x) := \frac{1}{\sqrt{2\pi}}e^{-x^2/2}, \quad x \in \mathbb{R},$$

where Z is standard normal.

The classical finite-time ruin probability is explicitly known:

$$(3) \quad \psi(u, T) := \mathbb{P}\left\{\sup_{0 \leq t \leq T} X(t) > u\right\} = \Psi\left(\frac{u + cT}{\sqrt{T}}\right) + e^{-2cu}\Psi\left(\frac{u - cT}{\sqrt{T}}\right).$$

The following elementary comparison records the ordering of the repeated-ruin functionals.

Proposition 2.1. *For every $u > 0$,*

$$(4) \quad \mathbb{P}\left\{\bigcap_{k=1}^{m-1} \{X(kh) > u\}\right\} \leq p_h^{\text{sl}}(u, T) \leq p_h^{\text{bl}}(u, T) \leq \mathbb{P}\left\{\sup_{0 \leq t \leq h} X(t) > u\right\}.$$

Remark 2.2. The first inequality in (4) connects the sliding-window event to a finite-dimensional Gaussian probability. The last inequality shows that both repeated-window probabilities

are bounded above by the classical ruin probability over the first reporting period, giving a sharper bound than the classical ruin probability over the full horizon.

For $m \geq 2$, define

$$(5) \quad G_m(h, c) := \mathbb{P} \left\{ \bigcap_{k=1}^{m-2} \left\{ \sup_{t \in [kh, (k+1)h]} (B(t) - ct) > 0 \right\} \right\}$$

with the convention $G_2(h, c) := 1$. This is the finite-level crossing probability for the limiting post-hit process. The interval $[0, h]$ imposes no condition in this constant, since Brownian motion with drift has positive values arbitrarily close to zero almost surely.

Theorem 2.3. *As $u \rightarrow \infty$,*

$$(6) \quad p_h^{\text{bl}}(u, T) \sim 2G_m(h, c) \Psi \left(\frac{u + ch}{\sqrt{h}} \right).$$

For $m \geq 2$, define the finite-level sliding constant

$$(7) \quad G_m^{\text{sl}}(h, c) := \mathbb{P} \left\{ \inf_{0 \leq r \leq (m-2)h} \sup_{r \leq s \leq r+h} (B(s) - cs) > 0 \right\}.$$

For $m = 2$, the infimum in (7) is over the singleton $\{0\}$, and hence $G_2^{\text{sl}}(h, c) = 1$ by the immediate upward oscillations of Brownian motion.

For $m \geq 2$, define the finite-dimensional grid constant

$$(8) \quad J_m(h, c) := \mathbb{P} \left\{ \bigcap_{j=1}^{m-2} \{B(jh) - cjh > 0\} \right\},$$

with the convention $J_2(h, c) := 1$. For $m \geq 3$, the defining Gaussian vector is non-degenerate, so $J_m(h, c) > 0$.

Theorem 2.4. *As $u \rightarrow \infty$,*

$$(9) \quad p_h^{\text{sl}}(u, T) \sim 2G_m^{\text{sl}}(h, c) \Psi \left(\frac{u + ch}{\sqrt{h}} \right).$$

The finite-level constants satisfy

$$(10) \quad 0 < J_m(h, c) \leq G_m^{\text{sl}}(h, c) \leq G_m(h, c) \leq 1, \quad m \geq 2.$$

Remark 2.5. The asymptotics in [Theorems 2.3](#) and [2.4](#) share the classical first-block ruin cost $2\Psi((u + ch)/\sqrt{h})$. Their remaining factors, G_m and G_m^{sl} , describe the subsequent finite-level requirements; neither probability is a product of one-block ruin probabilities. For $m = 2$, both factors equal one. Conditional on first-block ruin, the first hitting time τ_u of u approaches h . Measured from this hit, the second block is $[d_u, d_u + h]$, where $d_u = h - \tau_u$ tends to zero in conditional probability. Its crossing probability therefore tends to $\mathbb{P}\{\sup_{0 \leq s \leq h} (B(s) - cs) > 0\} = 1$. The sliding requirement has the same limit by right-continuity of its post-hit survival probability; see the proof of [Theorem 2.4](#). Consequently,

$$p_h^{\text{sl}}(u, 2h) \sim p_h^{\text{bl}}(u, 2h) \sim \psi(u, h).$$

The constant in (7) is a finite-horizon Parisian-type survival probability. Indeed, if

$$\tau_h^- := \inf \left\{ t \geq h : \sup_{t-h \leq s \leq t} X(s) \leq 0 \right\},$$

then

$$(11) \quad G_m^{\text{sl}}(h, c) = \mathbb{P}\{\tau_h^- > (m-1)h\}.$$

3. EXTENSIONS AND DISCUSSION

3.1. Simultaneous block ruin. Let $\rho \in (-1, 1)$, let B_1 and B_2 be independent copies of B , and put

$$(12) \quad W_1(t) = B_1(t), \quad W_2(t) = \rho B_1(t) + \sqrt{1 - \rho^2} B_2(t), \quad t \geq 0.$$

Fix $c_1, c_2 > 0$ and $a \in (\rho, 1]$, and define

$$X_i(t) = W_i(t) - c_i t, \quad i = 1, 2.$$

Set

$$(13) \quad p_{h,\rho}^{\text{ubl}}(u, T) := \mathbb{P} \left\{ \bigcap_{k=1}^m \{ \exists t \in [(k-1)h, kh] : X_1(t) > u, X_2(t) > au \} \right\}.$$

Thus (13) requires simultaneous ruin at least once in every reporting block. The insurance interpretation with two positive initial reserves corresponds to $a > 0$; the mathematical result below also covers $a \leq 0$ when $\rho < a$.

Let

$$\varphi_\rho(x, y) := \frac{1}{2\pi\sqrt{1-\rho^2}} \exp \left\{ -\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)} \right\}$$

be the density of $(W_1(1), W_2(1))$. Define

$$(14) \quad \lambda_1 = \frac{1 - a\rho}{1 - \rho^2}, \quad \lambda_2 = \frac{a - \rho}{1 - \rho^2},$$

and the simultaneous-ruin constant

$$(15) \quad C_{a,\rho} := \int_{\mathbb{R}^2} \mathbb{P} \{ \exists t \geq 0 : W_1(t) - t > x, W_2(t) - at > y \} e^{\lambda_1 x + \lambda_2 y} dx dy.$$

This is the constant in [7, Theorem 2.1(i)]. Its finiteness and strict positivity are proved directly below, together with the simultaneous block-ruin asymptotic. Finally, for $m \geq 3$, put

$$(16) \quad \mathcal{H}_m(h, c_1, c_2, \rho) := \mathbb{P} \left\{ \bigcap_{k=1}^{m-2} \{ \exists t \in [kh, (k+1)h] : W_1(t) - c_1 t > 0, W_2(t) - c_2 t > 0 \} \right\}$$

and set $\mathcal{H}_2(h, c_1, c_2, \rho) = 1$. For $m \geq 3$, requiring both coordinates to be positive at each time $h, \dots, (m-2)h$ gives a positive-probability Gaussian grid event contained in (16). Thus $\mathcal{H}_m(h, c_1, c_2, \rho) > 0$.

Theorem 3.1. *For the model above, as $u \rightarrow \infty$,*

$$(17) \quad p_{h,\rho}^{\text{ubl}}(u, T) \sim \mathcal{H}_m(h, c_1, c_2, \rho) C_{a,\rho} \frac{h}{u^2} \varphi_\rho \left(\frac{u + c_1 h}{\sqrt{h}}, \frac{au + c_2 h}{\sqrt{h}} \right).$$

The restriction $a > \rho$ isolates the bivariate binding regime, in which both coordinates contribute to the leading Gaussian cost and the constant $C_{a,\rho}$ appears.

3.2. Brownian-perturbed compound-Poisson risk model. To incorporate individual claims alongside continuous fluctuations, we consider the classical compound-Poisson risk process perturbed by Brownian motion, written here in aggregate-loss form; see [10, 8]. In this subsection, let

$$S(t) = \sum_{j=1}^{N(t)} Y_j, \quad X(t) = \sigma B(t) + S(t) - ct, \quad t \geq 0,$$

where N is a Poisson process with rate $\lambda > 0$, the claim sizes Y_j are i.i.d. and non-negative with distribution F , and N is independent of the claims. Assume that B is independent of S , $\sigma \geq 0$, and $c > 0$. The reserve is $R_u(t) = u - X(t)$; the Brownian component allows continuous fluctuations between claim arrivals.

Write $\bar{F}(u) = \mathbb{P}\{Y_1 > u\}$, $u \geq 0$, and assume that F is subexponential. Define p_h^{bl} and p_h^{sl} by (1) and (2), now using this X . No finite-mean or net-profit condition is needed for the fixed-horizon result below.

Proposition 3.2. *For the model above, as $u \rightarrow \infty$,*

$$p_h^{\text{sl}}(u, T) \sim p_h^{\text{bl}}(u, T) \sim \lambda h \bar{F}(u).$$

3.3. Discussion. The scalar Brownian proofs share a first-hitting-time argument. Conditional on a first-block exceedance, the first hit of u approaches h ; the strong Markov property then separates the first-block ruin cost from the remaining finite-level event. For blocks this event has probability $G_m(h, c)$, whereas sliding windows give the Parisian-type survival probability $G_m^{\text{sl}}(h, c)$. Thus the full horizon enters through these constants.

For simultaneous bivariate ruin, conditioning at the high endpoint at time h gives the same factorisation into a first-block cost and the later-block factor $\mathcal{H}_m(h, c_1, c_2, \rho)$. The binding regime changes the first-block Gaussian cost, which involves $C_{a, \rho}$ and a bivariate normal density.

In the perturbed model, the subexponential tail of the total claims in the first block determines the asymptotics. Once these claims keep the process above u throughout $[h, T]$, every length- h window is covered. Bounds using the Brownian extrema show that the perturbation changes neither the order nor the constant λh . In all three models, dependence across windows is essential: the repeated-ruin probability is not a product of one-window probabilities.

4. PROOFS

We first establish first-passage localization and continuity of simultaneous-crossing probabilities. We use the usual augmented Brownian filtrations throughout. For the scalar proofs, let $X(t) = B(t) - ct$, with $c > 0$, and define the first hitting time

$$\tau_u := \inf\{t \geq 0 : X(t) = u\}.$$

In expectations below, products containing $\mathbf{1}_{\{\tau_u < h\}}$ are understood to be zero on $\{\tau_u \geq h\}$. Immediate upward oscillations after τ_u imply that $\{\tau_u < h\} = \{\sup_{0 \leq t \leq h} X(t) > u\}$ almost surely.

Lemma 4.1. *As $u \rightarrow \infty$,*

$$(18) \quad \mathbb{P}\{\tau_u < h\} \sim 2\Psi\left(\frac{u + ch}{\sqrt{h}}\right), \quad \tau_u \mid \{\tau_u < h\} \longrightarrow h \quad \text{in probability.}$$

Consequently, if $q : (0, h] \rightarrow \mathbb{R}$ is bounded and measurable and $q(d) \rightarrow q_0$ as $d \downarrow 0$, then

$$(19) \quad \frac{\mathbb{E}\{\mathbf{1}_{\{\tau_u < h\}} q(h - \tau_u)\}}{\mathbb{P}\{\tau_u < h\}} \longrightarrow q_0.$$

Proof. The first assertion follows from (3) and Mills' ratio. For every $\delta \in (0, h)$, the same formula gives

$$\mathbb{P}\{\tau_u \leq h - \delta \mid \tau_u < h\} = \frac{\psi(u, h - \delta)}{\psi(u, h)} \longrightarrow 0,$$

because the numerator has a strictly larger Gaussian exponential cost. This proves the conditional convergence. Boundedness of q and its limit at zero then give (19). $\square$

For the following lemma, let $Y_i(t) = W_i(t) - d_i t$, $i = 1, 2$, where $d_1, d_2 \in \mathbb{R}$ and $\rho \in (-1, 1)$.

Lemma 4.2. *If $0 < r < s$, then*

$$(20) \quad \mathbb{P} \left\{ \sup_{t \in [r, s]} \min\{Y_1(t), Y_2(t)\} = 0 \right\} = 0.$$

Consequently, if $\eta_n = (\eta_{n,1}, \eta_{n,2}) \rightarrow (0, 0)$ and $I_j = [r_j, s_j]$, $j = 1, \dots, \ell$, with $r_j > 0$, then

$$(21) \quad \mathbb{P} \left\{ \bigcap_{j=1}^{\ell} \{ \exists t \in I_j : Y_1(t) > \eta_{n,1}, Y_2(t) > \eta_{n,2} \} \right\} \\ \longrightarrow \mathbb{P} \left\{ \bigcap_{j=1}^{\ell} \{ \exists t \in I_j : Y_1(t) > 0, Y_2(t) > 0 \} \right\}$$

as $n \rightarrow \infty$. Moreover, for every $s > 0$,

$$(22) \quad \mathbb{P} \{ \exists t \in [0, s] : Y_1(t) > \eta_{n,1}, Y_2(t) > \eta_{n,2} \} \longrightarrow 1$$

as $n \rightarrow \infty$.

Proof. For $I = [r, s]$, set

$$F_I(\eta) := \sup_{t \in I} \min\{Y_1(t) - \eta_1, Y_2(t) - \eta_2\}.$$

Then

$$(23) \quad |F_I(\eta) - F_I(\vartheta)| \leq \|\eta - \vartheta\|_{\infty}.$$

To prove (20), condition on the increments of (W_1, W_2) after time r . Given these increments, define

$$g_i(v) := Y_i(r + v) - Y_i(r), \quad 0 \leq v \leq s - r, \quad i = 1, 2,$$

and set

$$z = (Y_1(r), Y_2(r)).$$

Thus, under the conditioning, $g = (g_1, g_2)$ is a deterministic continuous function. Moreover,

$$F_I(0) = f_g(z),$$

where

$$f_g(z) := \sup_{0 \leq v \leq s-r} \min\{z_1 + g_1(v), z_2 + g_2(v)\}.$$

Since

$$f_g(z + t(1, 1)) = f_g(z) + t, \quad t \in \mathbb{R},$$

the zero set of f_g meets each line parallel to $(1, 1)$ in at most one point and therefore has two-dimensional Lebesgue measure zero. The vector $(Y_1(r), Y_2(r))$ is independent of the post- r increments and has a non-degenerate Gaussian density. Hence, conditional on the post- r increments,

$$\mathbb{P}(F_I(0) = 0 \mid \text{post-}r \text{ increments}) = 0.$$

Integrating over the post- r increments proves (20). The convergence in (21) now follows from (23), the no-atom statement, and dominated convergence.

It remains to consider an interval starting at zero. A non-degenerate two-dimensional Brownian motion is regular at the origin for every nonempty open cone. Indeed, in the zero-drift case the events

$$A_n := \{\exists t \in (0, 1/n) : (W_1(t), W_2(t)) \in (0, \infty)^2\}$$

have the same strictly positive probability by Brownian scaling. Their intersection belongs to the Brownian germ sigma-field, so Blumenthal's zero-one law shows that it has probability one. Adding a deterministic linear drift does not change this conclusion, because the drifted and drift-free laws are equivalent on every compact interval. Thus $F_{[0,s]}(0) > 0$ almost surely, and (23) gives (22). $\square$

Proof of Proposition 2.1. On the grid event in (4), the finite minimum

$$\eta := \min_{1 \leq k \leq m-1} \{X(kh) - u\}$$

is strictly positive. Every interval of length h in $[0, T]$ contains one of the points $h, 2h, \dots, (m-1)h$, so its supremum is at least $u + \eta$. This proves the first inequality. Each reporting block is a particular sliding window, proving the second inequality; retaining only the first block proves the last one. $\square$

Proof of Theorem 2.3. Let $Y(t) = \tilde{B}(t) - ct$, where $\tilde{B}$ is a standard Brownian motion, and define, for $0 \leq d \leq h$,

$$K_m(d) := \mathbb{P} \left\{ \bigcap_{k=0}^{m-2} \left\{ \sup_{d+kh \leq s \leq d+(k+1)h} Y(s) > 0 \right\} \right\}.$$

On $\{\tau_u < h\}$, the first block contains an exceedance almost surely. The strong Markov property at τ_u therefore gives the exact identity

$$(24) \quad p_h^{\text{bl}}(u, T) = \mathbb{E} \{ \mathbf{1}_{\{\tau_u < h\}} K_m(h - \tau_u) \}.$$

Indeed, relative to τ_u , the remaining reporting blocks are $[d+kh, d+(k+1)h]$, $k = 0, \dots, m-2$, where $d = h - \tau_u$.

We claim that

$$(25) \quad K_m(d) \longrightarrow K_m(0) = G_m(h, c), \quad d \downarrow 0.$$

Continuity of the paths implies, for each k ,

$$\sup_{d+kh \leq s \leq d+(k+1)h} Y(s) \longrightarrow M_k := \sup_{kh \leq s \leq (k+1)h} Y(s).$$

For $k = 0$, immediate upward oscillations give $M_0 > 0$ almost surely. For $k \geq 1$, write

$$M_k = Y(kh) + \sup_{0 \leq s \leq h} \{Y(kh + s) - Y(kh)\}.$$

The two terms are independent, and $Y(kh)$ has a Gaussian density. Thus $\mathbb{P}\{M_k = 0\} = 0$. The indicators of the finite intersections converge almost surely, proving (25) by dominated convergence. The equality $K_m(0) = G_m(h, c)$ follows by discarding the almost sure condition $M_0 > 0$.

Apply Lemma 4.1 with $q = K_m$ to (24). Equations (18) and (25) give

$$p_h^{\text{bl}}(u, T) \sim 2G_m(h, c) \Psi \left(\frac{u + ch}{\sqrt{h}} \right).$$

Here $G_m(h, c) \geq J_m(h, c) > 0$, since the grid event defining $J_m(h, c)$ implies every required block crossing. $\square$

Proof of Theorem 2.4. For $R \geq h$, put

$$H(R) := \mathbb{P}\{\tau_h^- > R\} = \mathbb{P}\{Q_{R,h} > 0\}, \quad Q_{R,h} := \inf_{0 \leq r \leq R-h} \sup_{r \leq s \leq r+h} Y(s),$$

where $Y(t) = \tilde{B}(t) - ct$. The second equality follows from continuity of the moving-window supremum and compactness of the range of window starts. In particular,

$$H(T-h) = G_m^{\text{sl}}(h, c).$$

As a survival function, H is right-continuous.

We first establish the exact identity

$$(26) \quad p_h^{\text{sl}}(u, T) = \mathbb{E}\{\mathbf{1}_{\{\tau_u < h\}} H(T - \tau_u)\}.$$

The sliding event requires $\tau_u < h$. On this event, use the strong Markov property to write $X(\tau_u + s) - u = Y(s)$. The windows starting at or after τ_u are covered with a uniform positive margin precisely when $Q_{T-\tau_u, h} > 0$. Almost surely, immediate upward oscillations also give some $s_0 \in (0, h - \tau_u)$ with $Y(s_0) > 0$. The point $\tau_u + s_0$ belongs to every window $[t, t+h]$ with $0 \leq t \leq \tau_u$. Hence, when $Q_{T-\tau_u, h} > 0$,

$$\inf_{0 \leq t \leq T-h} \sup_{t \leq s \leq t+h} X(s) - u \geq \min\{Y(s_0), Q_{T-\tau_u, h}\} > 0.$$

This proves both directions of the event equivalence and then (26) by conditional expectation.

The horizon $T - h + d$ approaches $T - h$ from above, so right-continuity gives

$$H(T - h + d) \longrightarrow H(T - h) = G_m^{\text{sl}}(h, c), \quad d \downarrow 0.$$

Apply Lemma 4.1 with $q(d) = H(T - h + d)$ to obtain

$$(27) \quad \frac{p_h^{\text{sl}}(u, T)}{\Psi((u + ch)/\sqrt{h})} \longrightarrow 2G_m^{\text{sl}}(h, c).$$

For the comparison of the constants, all three equal one when $m = 2$. If $m \geq 3$, on the event defining $J_m(h, c)$, the finite minimum

$$\eta := \min_{1 \leq j \leq m-2} Y(jh)$$

is positive. Every length- h window in $[0, (m-1)h]$ contains one of these grid points, so $Q_{(m-1)h, h} \geq \eta > 0$. Thus $J_m(h, c) \leq G_m^{\text{sl}}(h, c)$. Evaluating the sliding condition at $r = kh$, $k = 1, \dots, m-2$, gives $G_m^{\text{sl}}(h, c) \leq G_m(h, c) \leq 1$. Together with $J_m(h, c) > 0$, this proves (10) and makes (27) the claimed asymptotic equivalence. $\square$

Remark 4.3 (Exact first-passage integral). Differentiating (3) with respect to the horizon gives the first-passage density

$$\mathbb{P}\{\tau_u \in dt\} = \frac{u}{\sqrt{2\pi t^3}} \exp\left\{-\frac{(u + ct)^2}{2t}\right\} dt, \quad t > 0.$$

Consequently, (26) also yields

$$(28) \quad p_h^{\text{sl}}(u, T) = \int_0^h H(T-t) \frac{u}{\sqrt{2\pi t^3}} \exp\left\{-\frac{(u + ct)^2}{2t}\right\} dt.$$

Proof of Theorem 3.1. Write

$$\mathbf{b} = (1, a)^\top, \quad \mathbf{c} = (c_1, c_2)^\top, \quad \Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \quad \boldsymbol{\lambda} = \Sigma^{-1} \mathbf{b}.$$

Both components of $\boldsymbol{\lambda}$ are positive in the stated range. Put

$$\kappa_{a,\rho} = \mathbf{b}^\top \Sigma^{-1} \mathbf{b} = \frac{1 - 2a\rho + a^2}{1 - \rho^2} > 0, \quad D_u = u^{-2} \varphi_\rho \left(\frac{u + c_1 h}{\sqrt{h}}, \frac{au + c_2 h}{\sqrt{h}} \right).$$

We first verify directly that $0 < C_{a,\rho} < \infty$. Let $Z_i(t) = W_i(t) - b_i t$, and, for integers $n \geq 0$, set

$$M_{n,i} = \sup_{n \leq t \leq n+1} Z_i(t), \quad S_{n,i} = \sup_{0 \leq t \leq 1} \{W_i(n+t) - W_i(n) - b_i t\}.$$

Then $M_{n,i} = Z_i(n) + S_{n,i}$, and the vector $(S_{n,1}, S_{n,2})$ is independent of $(Z_1(n), Z_2(n))$ and has the same law for every n . The reflection principle gives all positive exponential moments of $\sup_{t \leq 1} W_i(t)$. Since $S_{0,i} \leq \sup_{t \leq 1} W_i(t) + |b_i|$, Cauchy–Schwarz shows that

$$J := \mathbb{E} \{ e^{\lambda_1 S_{0,1} + \lambda_2 S_{0,2}} \} < \infty.$$

Moreover, $\boldsymbol{\lambda}^\top \Sigma \boldsymbol{\lambda} = \boldsymbol{\lambda}^\top \mathbf{b} = \kappa_{a,\rho}$, so

$$\mathbb{E} \{ e^{\lambda_1 M_{n,1} + \lambda_2 M_{n,2}} \} = J e^{-n \kappa_{a,\rho}/2}.$$

A union bound over the unit intervals and Tonelli's theorem therefore give

$$C_{a,\rho} \leq \sum_{n=0}^{\infty} \int_{\mathbb{R}^2} \mathbb{P} \{ M_{n,1} > x, M_{n,2} > y \} e^{\lambda_1 x + \lambda_2 y} dx dy = \frac{J}{\lambda_1 \lambda_2} \sum_{n=0}^{\infty} e^{-n \kappa_{a,\rho}/2} < \infty.$$

The negative quadrant contributes $1/(\lambda_1 \lambda_2)$ already at time zero, proving strict positivity.

We next localize the first crossing near h . For $0 < r \leq h$, let

$$A_u(r) = \{ \exists t \in [0, r] : X_i(t) > b_i u, i = 1, 2 \}, \quad \tau_u = \inf \{ t \geq 0 : X_i(t) > b_i u, i = 1, 2 \}.$$

This entrance time into an open set is a stopping time. By continuity, $A_u(r) = \{ \tau_u < r \}$, and on $\{ \tau_u < \infty \}$ both $X_i(\tau_u) - b_i u$ are nonnegative. Define

$$q_h = \mathbb{P} \{ W_i(1) > c_i \sqrt{h}, i = 1, 2 \} > 0.$$

On $\{ \tau_u < r \}$, the strong Markov property, Brownian scaling, and $0 < r - \tau_u \leq h$ imply

$$\mathbb{P} \{ X_i(r) > b_i u, i = 1, 2 \mid \mathcal{F}_{\tau_u} \} \geq \mathbb{P} \{ W_i(1) > c_i \sqrt{r - \tau_u}, i = 1, 2 \} \geq q_h.$$

Here the middle probability is evaluated at the $\mathcal{F}_{\tau_u}$ -measurable remaining time. Consequently,

$$(29) \quad \mathbb{P} \{ A_u(r) \} \leq q_h^{-1} \mathbb{P} \{ X_i(r) > b_i u, i = 1, 2 \}.$$

For $r \in [h/2, h]$, Gaussian integration with endpoint offsets $\mathbf{z}/u$, $\mathbf{z} = (z_1, z_2)^\top \geq 0$, gives

$$\begin{aligned} & \mathbb{P} \{ X_i(r) > b_i u, i = 1, 2 \} \\ &= \frac{1}{ru^2} \varphi_\rho \left(\frac{u + c_1 r}{\sqrt{r}}, \frac{au + c_2 r}{\sqrt{r}} \right) \int_{[0, \infty)^2} \exp \left\{ -\frac{\boldsymbol{\lambda}^\top \mathbf{z}}{r} - \frac{\mathbf{c}^\top \Sigma^{-1} \mathbf{z}}{u} - \frac{\mathbf{z}^\top \Sigma^{-1} \mathbf{z}}{2ru^2} \right\} dz_1 dz_2. \end{aligned}$$

For all sufficiently large u , the integrand is bounded, uniformly in $r \in [h/2, h]$, by

$$\exp \left\{ -\sum_{i=1}^2 \frac{\lambda_i z_i}{2h} \right\}.$$

Its difference from $\exp \{ -\boldsymbol{\lambda}^\top \mathbf{z}/r \}$ tends to zero uniformly in r for each fixed $\mathbf{z}$. Dominated convergence applied to this supremum of differences shows that the integral equals $r^2/(\lambda_1 \lambda_2) + o(1)$, uniformly in r . Thus

$$(30) \quad \mathbb{P} \{ X_i(r) > b_i u, i = 1, 2 \} \sim \frac{r}{\lambda_1 \lambda_2 u^2} \varphi_\rho \left(\frac{u + c_1 r}{\sqrt{r}}, \frac{au + c_2 r}{\sqrt{r}} \right)$$

uniformly for $r \in [h/2, h]$.

Fix $K > 0$, put $r_u = h - K/u^2$, and let

$$A_u^{(K)}(h) = \{\exists t \in [r_u, h] : X_i(t) > b_i u, \ i = 1, 2\}.$$

The ratio of the Gaussian densities at times r_u and h is

$$\begin{aligned} & \frac{\varphi_\rho((u + c_1 r_u)/\sqrt{r_u}, (au + c_2 r_u)/\sqrt{r_u})}{\varphi_\rho((u + c_1 h)/\sqrt{h}, (au + c_2 h)/\sqrt{h})} \\ &= \exp\left\{-\frac{\kappa_{a,\rho} u^2}{2} \left(\frac{1}{r_u} - \frac{1}{h}\right) - \frac{\mathbf{c}^\top \Sigma^{-1} \mathbf{c}}{2} (r_u - h)\right\} \\ &\longrightarrow e^{-\kappa_{a,\rho} K/(2h^2)}. \end{aligned}$$

Since $A_u(h) \setminus A_u^{(K)}(h) \subseteq A_u(r_u)$, (29) and (30) imply

$$(31) \quad \limsup_{u \rightarrow \infty} \frac{\mathbb{P}\{A_u(h) \setminus A_u^{(K)}(h)\}}{D_u} \leq \frac{h}{q_h \lambda_1 \lambda_2} e^{-\kappa_{a,\rho} K/(2h^2)}.$$

Let E_u be the block event in (13), and set $E_u^{(K)} := E_u \cap A_u^{(K)}(h)$. Thus the crossing required in the first block is restricted to $[r_u, h]$; earlier crossings are allowed. Condition on $X_i(h) = b_i u + z_i/u$, where $(z_1, z_2) = (x, y)$, and set

$$\hat{X}_{u,i}^{x,y}(s) = u\{X_i(h - s/u^2) - b_i u\}, \quad 0 \leq s \leq K.$$

The conditional mean and covariance are

$$\begin{aligned} \mathbb{E}\{\hat{X}_{u,i}^{x,y}(s)\} &= z_i \left(1 - \frac{s}{hu^2}\right) - \frac{b_i s}{h}, \\ \text{Cov}\left(\hat{X}_{u,i}^{x,y}(s), \hat{X}_{u,j}^{x,y}(t)\right) &= \Sigma_{ij} \left(\min(s, t) - \frac{st}{hu^2}\right). \end{aligned}$$

For $L = hu^2$, the centered bridge has the law of $(W_i(s) - (s/L)W_i(L))_{i=1,2}$ on $[0, K]$. Its correction to W tends uniformly to zero in probability because $(K/L)\|W(L)\| \rightarrow 0$ in probability. Hence

$$(32) \quad (\hat{X}_{u,1}^{x,y}, \hat{X}_{u,2}^{x,y}) \Rightarrow (W_1(s) + x - s/h, W_2(s) + y - as/h)_{s \in [0, K]}$$

in $C([0, K], \mathbb{R}^2)$. Write

$$\begin{aligned} p_{u,K}(x, y) &= \mathbb{P}\{A_u^{(K)}(h) \mid X_1(h) = u + x/u, X_2(h) = au + y/u\}, \\ p_K(x, y) &= \mathbb{P}\{\exists s \in [0, K] : W_1(s) + x - s/h > 0, W_2(s) + y - as/h > 0\}. \end{aligned}$$

For a continuous path w , the functional

$$F_w(x, y) = \sup_{0 \leq s \leq K} \min\{w_1(s) + x - s/h, w_2(s) + y - as/h\}$$

is continuous in the path and satisfies $F_w(x + t, y + t) = F_w(x, y) + t$. Its zero set is therefore Lebesgue-null by integration along lines parallel to $(1, 1)$. Fubini's theorem gives $\mathbb{P}\{F_W(x, y) = 0\} = 0$ for almost every (x, y) . The continuous mapping theorem and (32) yield

$$(33) \quad p_{u,K}(x, y) \longrightarrow p_K(x, y) \quad \text{for Lebesgue-almost every } (x, y).$$

Let $\tilde{Y}_i(t) = W_i(h + t) - W_i(h) - c_i t$. Given the endpoint, the probability of the remaining blocks is

$$q_u(x, y) = \mathbb{P}\left\{\bigcap_{\ell=0}^{m-2} \left\{\exists t \in [\ell h, (\ell + 1)h] : \tilde{Y}_1(t) > -x/u, \tilde{Y}_2(t) > -y/u\right\}\right\}.$$

The $\ell = 0$ event has probability tending to one by (22); the intersection for $\ell \geq 1$ converges by (21). For $m = 2$ the latter intersection is empty. Thus, for every fixed (x, y) ,

$$(34) \quad q_u(x, y) \longrightarrow \mathcal{H}_m(h, c_1, c_2, \rho).$$

The endpoint density is

$$f_h(v_1, v_2) = h^{-1} \varphi_\rho \left((v_1 + c_1 h) / \sqrt{h}, (v_2 + c_2 h) / \sqrt{h} \right),$$

and its ratio $R_u(x, y) = f_h(u + x/u, au + y/u) / f_h(u, au)$ equals

$$(35) \quad R_u(x, y) = \exp \left\{ -\frac{\lambda_1 x + \lambda_2 y}{h} - \frac{\mathbf{c}^\top \Sigma^{-1}(x, y)^\top}{u} - \frac{(x, y) \Sigma^{-1}(x, y)^\top}{2hu^2} \right\}.$$

The bridge before h and the increments after h are conditionally independent. Changing endpoint variables therefore gives

$$(36) \quad \frac{\mathbb{P}\{E_u^{(K)}\}}{D_u} = \frac{1}{h} \int_{\mathbb{R}^2} p_{u,K}(x, y) q_u(x, y) R_u(x, y) dx dy.$$

Here is an integrable bound uniform in all sufficiently large u , for this fixed K . Take $L = hu^2 \geq 2K$. Each centered bridge marginal $G_L(s) = B(s) - (s/L)B(L)$ satisfies, for $v \geq 0$,

$$\begin{aligned} \mathbb{P} \left\{ \sup_{s \leq K} G_L(s) > v \right\} &\leq \mathbb{P} \left\{ \sup_{s \leq K} B(s) > v/2 \right\} + \mathbb{P} \{|B(L)| > vL/(2K)\} \\ &\leq 2e^{-v^2/(8K)} + 2e^{-v^2 L/(8K^2)} \leq 4e^{-v^2/(8K)}, \end{aligned}$$

by reflection and the Gaussian tail bound. Put $z_i^- := \max(-z_i, 0)$ and $d_i = |b_i|K/h$. If $z_i < 0$, the conditional mean is at most $-z_i^-/2 + d_i$. A simultaneous crossing thus requires the corresponding centered bridge to exceed $z_i^-/2 - d_i$. Using $(r/2 - d)_+^2 \geq r^2/8 - d^2$, we obtain, also for $z_i \geq 0$,

$$p_{u,K}(x, y) \leq C_i e^{-(z_i^-)^2/(64K)}, \quad C_i = 4e^{d_i^2/(8K)}, \quad i = 1, 2.$$

Taking the geometric mean requires no independence and yields

$$p_{u,K}(x, y) \leq C_K \exp \left\{ -\frac{(x^-)^2 + (y^-)^2}{128K} \right\}, \quad C_K = \sqrt{C_1 C_2}.$$

Choose $0 < \varepsilon < \min(\lambda_1, \lambda_2)/(2h)$. The quadratic term in (35) is nonpositive, and, for all sufficiently large u ,

$$R_u(x, y) \leq \exp \left\{ -\frac{\lambda_1 x + \lambda_2 y}{h} + \varepsilon(|x| + |y|) \right\}.$$

Since $0 \leq q_u \leq 1$, an upper bound for the integrand in (36) is

$$C_K \prod_{i=1}^2 \exp \left\{ -\frac{\lambda_i}{2h} z_i^+ - \frac{(z_i^-)^2}{128K} + \left(\frac{\lambda_i}{h} + \varepsilon \right) z_i^- \right\}, \quad z_i^+ = \max(z_i, 0).$$

This is integrable on $\mathbb{R}^2$, with exponential decay on the positive half-axes and Gaussian decay on the negative half-axes. Dominated convergence, (33), and (34) now give

$$(37) \quad \frac{\mathbb{P}\{E_u^{(K)}\}}{D_u} \longrightarrow \mathcal{H}_m(h, c_1, c_2, \rho) L_K(h),$$

where

$$(38) \quad L_K(h) = \frac{1}{h} \int_{\mathbb{R}^2} p_K(x, y) e^{-(\lambda_1 x + \lambda_2 y)/h} dx dy.$$

Substituting $x = -hr$, $y = -hz$, $s = h^2t$, and applying Brownian scaling gives

$$L_K(h) = h \int_{\mathbb{R}^2} \mathbb{P}\{\exists t \in [0, K/h^2] : W_1(t) - t > r, W_2(t) - at > z\} e^{\lambda_1 r + \lambda_2 z} dr dz.$$

Monotone convergence and (15) show that

$$(39) \quad L_K(h) \uparrow hC_{a,\rho} \quad (K \rightarrow \infty).$$

Finally,

$$E_u^{(K)} \subseteq E_u \subseteq E_u^{(K)} \cup (A_u(h) \setminus A_u^{(K)}(h)).$$

Equations (31) and (37) therefore imply

$$\begin{aligned} \mathcal{H}_m(h, c_1, c_2, \rho) L_K(h) &\leq \liminf_{u \rightarrow \infty} \frac{\mathbb{P}\{E_u\}}{D_u} \leq \limsup_{u \rightarrow \infty} \frac{\mathbb{P}\{E_u\}}{D_u} \\ &\leq \mathcal{H}_m(h, c_1, c_2, \rho) L_K(h) + \frac{h}{q_h \lambda_1 \lambda_2} e^{-\kappa_{a,\rho} K / (2h^2)}. \end{aligned}$$

Letting $K \rightarrow \infty$ proves (17). □

Proof of Proposition 3.2. Put

$$Z_T := cT + \sigma \sup_{0 \leq s \leq T} |B(s)|.$$

Since the claims are non-negative,

$$(40) \quad \mathbb{P}\{S(h) - Z_T > u\} \leq p_h^{\text{sl}}(u, T) \leq p_h^{\text{bl}}(u, T) \leq \mathbb{P}\{S(h) + Z_T > u\}.$$

Indeed, on the event on the left, the margin $\eta = S(h) - Z_T - u$ is positive and $X(s) \geq u + \eta$ for every $s \in [h, T]$. Every length- h window meets $[h, T]$, so its supremum is at least $u + \eta$; this also verifies the strict sliding infimum. The middle inequality follows by selecting the reporting blocks. For the last inequality, $X(s) \leq S(h) + Z_T$ on $[0, h]$.

The compound-Poisson tail asymptotic [9, Theorem 3.37] gives

$$(41) \quad \mathbb{P}\{S(h) > u\} \sim \lambda h \bar{F}(u).$$

The variable Z_T is independent of $S(h)$ and has finite exponential moments. Since F is long-tailed,

$$\mathbb{P}\{\pm Z_T > u\} = o(\bar{F}(u))$$

for either choice of sign. Apply [9, Corollary 3.19], which permits distributions on the whole real line, with reference distribution F and summands $S(h)$ and $\pm Z_T$. Their limiting tail ratios are λh and zero, respectively, and hence

$$\mathbb{P}\{S(h) \pm Z_T > u\} \sim \lambda h \bar{F}(u).$$

The sandwich (40) proves the proposition. □

REFERENCES

- [1] K. Bisewski, K. Dębicki, and N. Kriukov, *Simultaneous ruin probability for multivariate Gaussian risk model*, Stochastic Process. Appl. **160** (2023), 386–408, [doi:10.1016/j.spa.2023.03.002](https://doi.org/10.1016/j.spa.2023.03.002).
- [2] I. Czarna and Z. Palmowski, *Ruin probability with Parisian delay for a spectrally negative Lévy risk process*, J. Appl. Probab. **48** (2011), no. 4, 984–1002, [doi:10.1239/jap/1324046014](https://doi.org/10.1239/jap/1324046014).
- [3] K. Dębicki, E. Hashorva, and L. Ji, *Parisian ruin of self-similar Gaussian risk processes*, J. Appl. Probab. **52** (2015), no. 3, 688–702, [doi:10.1239/jap/1445543840](https://doi.org/10.1239/jap/1445543840).
- [4] K. Dębicki, E. Hashorva, and L. Ji, *Parisian ruin over a finite-time horizon*, Sci. China Math. **59** (2016), no. 3, 557–572, [doi:10.1007/s11425-015-5073-6](https://doi.org/10.1007/s11425-015-5073-6).
- [5] K. Dębicki, E. Hashorva, L. Ji, and T. Rolski, *Extremal behavior of hitting a cone by correlated Brownian motion with drift*, Stochastic Process. Appl. **128** (2018), no. 12, 4171–4206, [doi:10.1016/j.spa.2018.02.002](https://doi.org/10.1016/j.spa.2018.02.002).

- [6] K. Dębicki, E. Hashorva, L. Ji, and K. Tabiś, *Extremes of vector-valued Gaussian processes: Exact asymptotics*, Stochastic Process. Appl. **125** (2015), no. 11, 4039–4065, [doi:10.1016/j.spa.2015.05.015](https://doi.org/10.1016/j.spa.2015.05.015).
- [7] K. Dębicki, E. Hashorva, and Z. Michna, *Simultaneous ruin probability for two-dimensional Brownian risk model*, J. Appl. Probab. **57** (2020), no. 2, 597–612, [doi:10.1017/jpr.2020.14](https://doi.org/10.1017/jpr.2020.14).
- [8] F. Dufresne and H. U. Gerber, *Risk theory for the compound Poisson process that is perturbed by diffusion*, Insurance Math. Econom. **10** (1991), no. 1, 51–59, [doi:10.1016/0167-6687\(91\)90023-Q](https://doi.org/10.1016/0167-6687(91)90023-Q).
- [9] S. Foss, D. Korshunov, and S. Zachary, *An Introduction to Heavy-Tailed and Subexponential Distributions*, 2nd ed., Springer, New York, 2013.
- [10] H. U. Gerber, *An extension of the renewal equation and its application in the collective theory of risk*, Scand. Actuar. J. **1970** (1970), no. 3–4, 205–210, [doi:10.1080/03461238.1970.10405664](https://doi.org/10.1080/03461238.1970.10405664).
- [11] P. Ievlev and S. Novikov, *Extremes of uncorrelated gamma-reflected Brownian motions with trend*, Scand. Actuar. J. **2026** (2026), no. 6, 630–656, [doi:10.1080/03461238.2025.2587662](https://doi.org/10.1080/03461238.2025.2587662).
- [12] G. Jasnovidov, *Approximation of ruin probability and ruin time in discrete Brownian risk models*, Scand. Actuar. J. **2020** (2020), no. 8, 718–735, [doi:10.1080/03461238.2020.1725911](https://doi.org/10.1080/03461238.2020.1725911).
- [13] R. Loeffen, I. Czarna, and Z. Palmowski, *Parisian ruin probability for spectrally negative Lévy processes*, Bernoulli **19** (2013), no. 2, 599–609, [doi:10.3150/11-BEJ404](https://doi.org/10.3150/11-BEJ404).
- [14] S. M. Novikov, *Parisian ruin of locally self-similar Gaussian processes*, Scand. Actuar. J. (2026), published online 28 May 2026, 1–27, [doi:10.1080/03461238.2026.2672546](https://doi.org/10.1080/03461238.2026.2672546).

KRZYSZTOF DĘBICKI, MATHEMATICAL INSTITUTE, UNIVERSITY OF WROCŁAW, PL. GRUNWALDZKI 2/4, 50-384 WROCŁAW, POLAND

Email address: Krzysztof.Debicki@math.uni.wroc.pl

ENKELEJD HASHORVA, DEPARTMENT OF ACTUARIAL SCIENCE, UNIVERSITY OF LAUSANNE, CHAMBERONNE 1015 LAUSANNE, SWITZERLAND

Email address: Enkelejd.Hashorva@unil.ch

ZBIGNIEW MICHNA, DEPARTMENT OF OPERATIONS RESEARCH AND BUSINESS INTELLIGENCE, WROCŁAW UNIVERSITY OF SCIENCE AND TECHNOLOGY, WYBRZEŻE STANISŁAWA WYSPIAŃSKIEGO 27, 50-370 WROCŁAW, POLAND

Email address: Zbigniew.Michna@pwr.edu.pl